CSA1006 – SOFTWARE ENGINEERING

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Unit 5 Assignment

AI/ML and Edge Computing in Modern Software Engineering

Introduction

Modern software engineering is changing rapidly because of technologies such as **Artificial Intelligence/Machine Learning (AI/ML)** and **Edge Computing**. AI/ML enables software systems to learn from data, identify patterns, automate decisions, and provide intelligent features. Edge Computing processes data closer to where it is generated instead of sending everything to a centralized cloud server.

Together, these technologies help developers build software that is more intelligent, responsive, efficient, and capable of handling large amounts of real-time data.

1. Role of AI/ML in Modern Software Engineering

Artificial Intelligence and Machine Learning allow software applications to perform tasks that traditionally required human decision-making or manual analysis.

How AI/ML transforms software engineering

1. Intelligent Automation

AI/ML can automate repetitive development and operational tasks such as code assistance, testing, bug detection, and software monitoring.

2. Predictive Analysis

Machine Learning models can analyze historical data and identify patterns. This can help predict system failures, user behavior, or performance problems before they become serious.

3. Automated Testing

AI/ML can assist in identifying potential defects and generating or prioritizing test cases. This can improve testing efficiency and help developers detect problems earlier.

4. Personalized Software

AI/ML can analyze user behavior and provide personalized recommendations or content. **5. Improved Decision-Making**

Software can use ML models to analyze large datasets and provide useful predictions or recommendations to users and organizations.

Practical Example – E-Commerce Recommendation System

An e-commerce application can use Machine Learning to analyze a customer's previous searches, purchases, and interactions.

Customer Activity

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Data Collection

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Machine Learning Model

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Analyze User Preferences

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Product Recommendations

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Personalized User Experience

For example, if a customer frequently searches for sports products, the system can recommend relevant sports shoes, clothing, or accessories.

Thus, AI/ML makes conventional software more **intelligent, adaptive, and personalized**.

2. Role of Edge Computing in Modern Software Engineering

Edge Computing is a computing approach in which data is processed closer to the device or location where it is generated rather than sending all data to a distant cloud server.

This is particularly useful for applications that require quick responses. **How Edge Computing transforms software engineering**

1. Reduced Latency

Since processing takes place close to the user or device, the time required to send data to a remote server and receive a response is reduced.

2. Faster Real-Time Processing

Edge computing is useful for applications that require immediate decisions, such as industrial monitoring, autonomous systems, and smart devices.

3. Reduced Network Traffic

Instead of sending every piece of raw data to the cloud, edge devices can process and filter data locally and send only important information.

4. Improved Reliability

Applications can continue performing certain operations even when connectivity to a central cloud service is temporarily unavailable.

5. Better Data Management

Sensitive or unnecessary raw data can be processed locally, reducing the amount of information that needs to be transmitted to centralized systems.

Practical Example – Smart Traffic Management

A smart-city traffic system can use cameras and edge devices installed near roads to analyze traffic conditions locally.

Traffic Cameras

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Edge Device

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Local Traffic Analysis

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Detect Traffic Congestion

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Traffic Signal Adjustment

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Cloud → Long-Term Analysis

Instead of sending continuous video streams to a remote cloud server, the edge device can analyze the video locally and send only useful information, such as vehicle counts or congestion alerts.

This enables faster responses and reduces network bandwidth usage.

3. AI/ML and Edge Computing Together

AI/ML and Edge Computing can also work together. An ML model can be deployed directly on an edge device so that the device can analyze data locally and make decisions quickly. For example, in a smart surveillance system:

Camera

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Edge Device

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AI/ML Model

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Analyze Video Locally

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Detect Relevant Event

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Send Alert / Result

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Cloud Storage & Analysis

This combination provides both **intelligence from AI/ML** and **low-latency processing from Edge Computing**.

4. Comparison

Aspect	**AI/ML**	**Edge Computing**
Main purpose	Makes software intelligent	Processes data closer to its source
Key benefit	Prediction and automation	Low latency and faster response
Typical use	Recommendations, prediction, automated testing	IoT, smart cities, real-time monitoring
Data processing	Uses data to learn patterns and make predictions	Processes data locally or near the source

Example Product recommendation system Smart traffic management

Conclusion

AI/ML and Edge Computing are transforming modern software engineering in different but complementary ways. **AI/ML** enables software to learn from data, automate tasks, make predictions, and provide personalized services. **Edge Computing** brings computation closer to data sources, reducing latency, network traffic, and dependence on centralized systems.

When combined, they can create software systems that are **intelligent, responsive, efficient, and suitable for real-time applications**. These technologies are therefore becoming important components of modern software development and deployment.